

Practice Problems (Function)

1. Create a Function to Calculate the Power of a Number a^b Without Using `pow()`

Write a function `int power(int a, int b)` to compute the power of `a` raised to `b` using a loop. For example, $2^3 = 8$. Handle edge cases like `b = 0` and `b < 0`.

2. Create a Function to Compute the Sum of All Proper Divisors of a Number

Define `int sumOfDivisors(int n)` that returns the sum of all divisors of `n`. Use a loop to find divisors of `n` by checking numbers up to `n/2`.

Sample Input: 10

Output: 8

// proper divisors 1 + 2 + 5 = 8

3. Write a Function to Find the Maximum of Three Numbers

Implement `int maxOfThree(int a, int b, int c)` to return the largest of three input numbers. Use conditional statements to compare the numbers.

4. Implement a Function to Determine Whether a Year Is a Leap Year

Create `int isLeapYear(int year)` that returns 1 if the year is a leap year and 0 otherwise. A year is a leap year if divisible by 4 but not by 100, unless also divisible by 400.

5. Create a Function to Calculate the LCM of Two Numbers

Define `int lcm(int a, int b)` to compute the LCM (least common multiple) of two integers `a` and `b`.

Sample Input: 15 25

Output: The LCM of 15 and 25 is 75.

6. Implement a Function That Converts a Temperature From Celsius to Fahrenheit and Vice Versa

Write `float convertTemp(float temp, char scale)` to convert temperatures between Celsius and Fahrenheit. Here `scale` parameter is your given temperature scale. Use the formulas:

- Celsius to Fahrenheit: $F = C \times 9/5 + 32$
- Fahrenheit to Celsius: $C = 5/9 \times (F - 32)$

7. Create a Function to Generate All Factors of a Number

Implement `void factors(int n)` to print all factors of `n`.

Sample Input: 12

Output: 1 2 3 4 6 12

8. Triangle Properties

Write a program to compute various properties of a triangle. Implement the following functions:

- `float calculatePerimeter(float a, float b, float c)` to calculate the perimeter of a triangle.
- `float calculateArea(float a, float b, float c)` to calculate the area using Heron's formula: $Area = \sqrt{s(s-a)(s-b)(s-c)}$
where `s` is the semi-perimeter $s = \frac{a+b+c}{2}$.
- `int isValidTriangle(float a, float b, float c)` to check if the three sides form a valid triangle (triangle inequality condition).

The main function should call `isValidTriangle` first, and if valid, proceed to calculate and display the perimeter and area using the respective functions.

9. Write a Function to Sort an Array of Numbers in Ascending Order

Implement `void sortArray(int arr[], int n)` to sort the input array of size `n` from smallest to largest.

10. Write a Function to Find the Sum of Elements in an Array

Define `int sumArray(int arr[], int n)` to calculate the sum of all elements in the input array of size `n`. Use a loop to iterate through the array.